Retinal longevity and the recovery measurement

What public data can answer now,
and what a future experiment must add.

Reproducible research + conditional funding package

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Four questions still need a matched observation

1	**Late identity**	Which age-positive human cells retain adult identity?
2	**Rare states**	What unusual states occur, and can we detect them reliably?
3	**Recovery**	What remains after withdrawal and longer follow-up?
4	**Continuity**	Can a common measured trajectory connect the late windows?

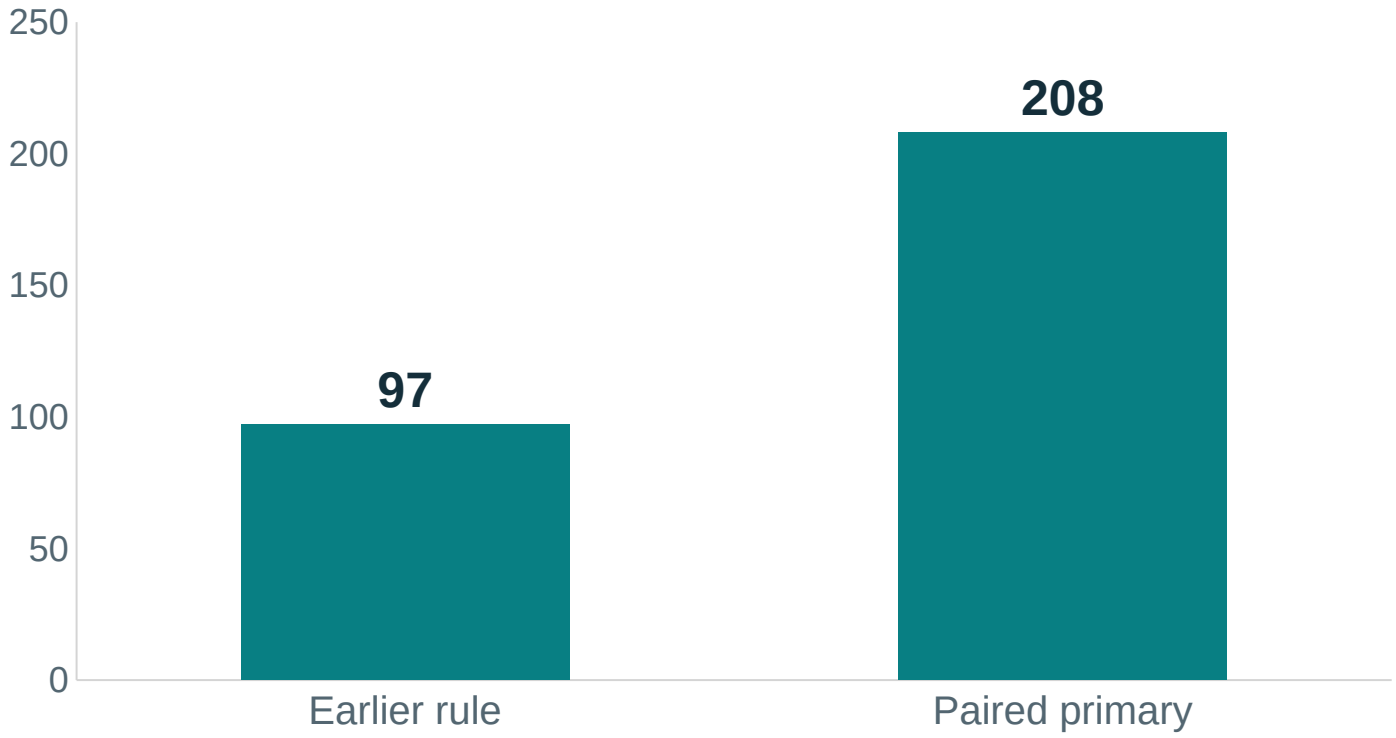
Current verdict for each: insufficient_signal

Existing data reduce what we need to collect

Human perturbations	36,514 cells; guide assignments reconstructed and audited
Human recovery	Three-donor SRSF1 diagnostic; earlier bulk recovery comparators
Ocular measurements	Mouse RPE sensitivity analysis; exact clock replay still pending
Observed withdrawal	Mouse liver factor-off data; useful within its own biological scope

Original experiments retain their authorship. Our work is reanalysis and evidence evaluation.

The paired-guide design changes the analysis



208

eligible comparisons

Both targeting guides count.

EZH2 moves toward zero. E2F3 fails the expected target-direction check.

Comparisons require ≥20 target and ≥20 matched NT cells. Reconstructed assignments.

The ocular endpoint needs to be named precisely

+0.663 to +0.694

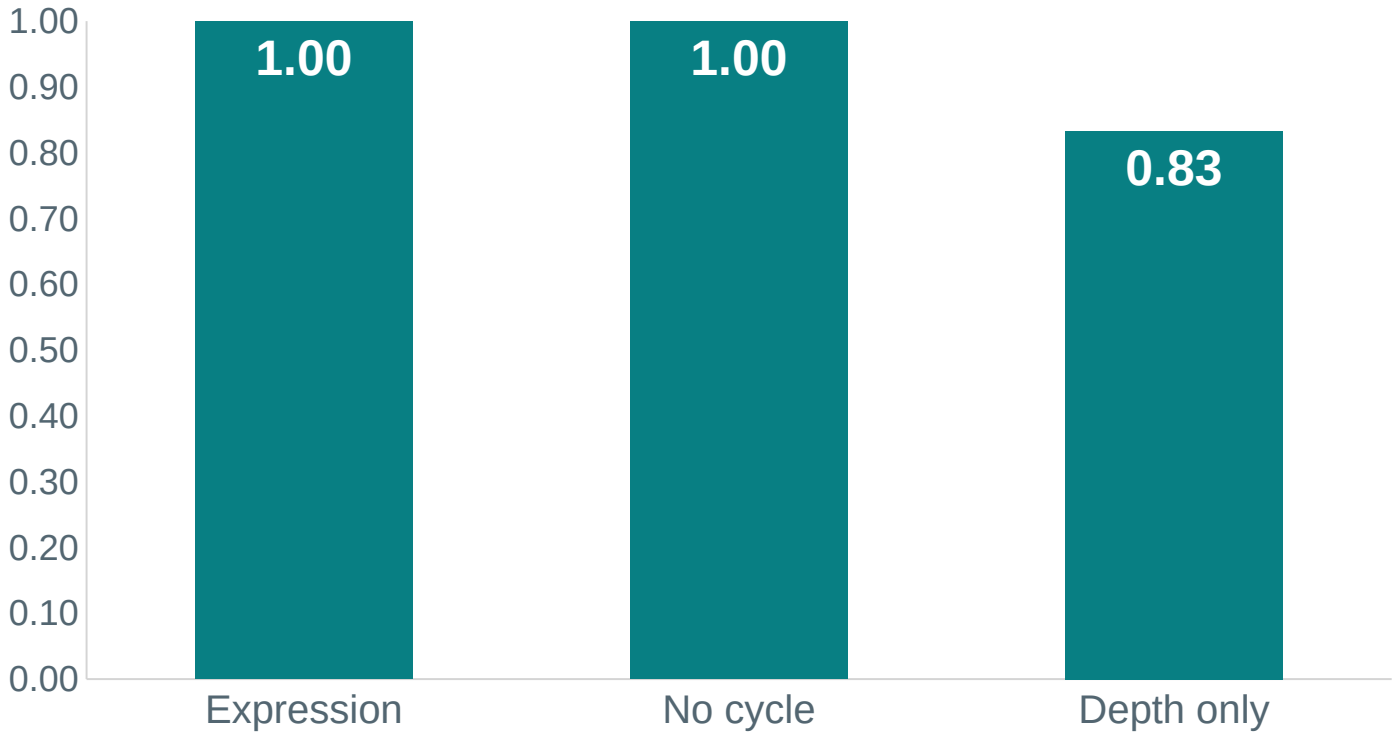
Broad expression correlation across six fixed sensitivities

Our calculation: gene-wise age and OSK contrasts.

Source study: a fitted transcriptomic clock with specific preprocessing.

Exact clock, imputation references and animal/cohort mapping remain prerequisites.

A model recognizes treatment in held-out donors



3 donors

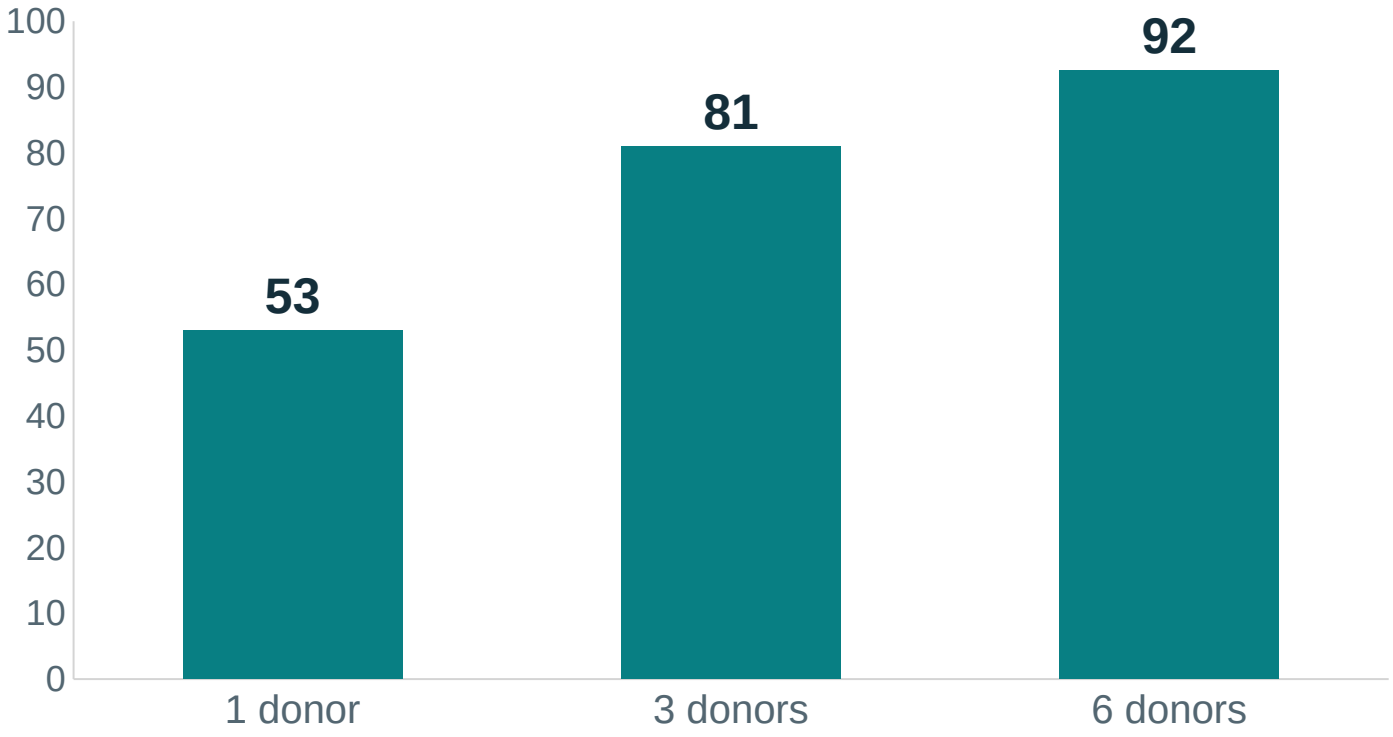
12 repeated libraries
Two donors train; one tests.

Treatment recognition

Experimental batch is missing.
No independent study test.

Mean held-donor AUROC. Balanced label controls average 0.50. Benefit and safety remain unmeasured.

More donors change rare-state detection



1,200 cells

Same total in each scenario

Assumed prevalence: 1%
Sensitivity: 80%
Latent donor correlation: 0.05

Synthetic only

Chance of ≥1 true detection (%). Hypothetical assumptions; detector false positives also need calibration.

Public withdrawal data exist — in another system

Mouse liver: one day on, one day off

16,042

deposited barcodes

2,288

source-QC sensitivity

2,284

paper-reported cells

Four-cell QC discrepancy remains. Early mouse liver data do not establish late human or retinal recovery.

A focused pilot keeps the three comparisons

Day	Reference	Induced / continued	Withdrawal
0	1 shared baseline	—	—
17	1	1 before branching	—
20	1	1	1
28	1	1	1
35	1	—	1

11 libraries per donor × 3 donors = 33

Proposed D17 branch. Feasibility and donor count require experimental partner review.

Fund the missing observation in stages

First	Recover metadata and assess compatible banked material.
33 libraries	Focused feasibility pilot; estimate donor variance and assay performance.
57 libraries	Full nine-timepoint ladder, including four earlier reference/induced pairs.
Then	Longer recovery and a separate retinal/RPE functional bridge, if justified.

Partner, calibrated endpoints and real quotes are pending. No wet-lab work has been purchased or arranged.

A public result others can inspect and extend

Code · source hashes · measured results · model card
Synthetic assumptions · failed paths · funding design

75 computational checks + 12 scientific failure-case tests

All four biological readouts remain open.

The most useful next contribution is the exact missing dataset or metadata.

github.com/SproutSeeds/retinal-longevity-controller-benchmarks